Rakan Eyad

Mechatronic Engineer

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Summary

Mechatronic engineering graduate (Honours, University of Sydney) with 14 months of industry experience on a prototype control system, plus two production web applications built solo.

Experience

System Engineer — ALBON Nov 2024 — Jan 2026

- Developed and integrated hardware, software, and mechanical components for a prototype control system: migrated the architecture from a microcontroller to a Raspberry Pi for real-time data processing, integrated sensor arrays, and engineered the system's electrical junction box to cut power consumption and keep the hardware reliable.

Engineering Intern — Escape The Room Sep 2021 — Nov 2021

- Designed mechanical puzzles and implemented engineering concepts in real-world, client-facing products.

Testing Technician Intern — CloudAk May 2021 — Aug 2021

- Supported server system testing and maintenance in data-centre environments.

Featured Projects

Conjure — Open source · Python

- Built a real-time computer-vision engine classifying 9 hand/face gestures via MediaPipe landmarks, rendering effects at 60 FPS with unit tests and CI.
- Profiled and optimized the render pipeline from 162 ms to 2.8 ms per frame (58x speed-up) through allocation discipline.

Planetary Rover RL Navigation — Thesis · Isaac Sim

- Developed an Honours thesis (78/100, Distinction) on autonomous rover navigation: an A* global planner coupled to a PPO local policy, trained in NVIDIA Isaac Sim with 64 parallel rovers on Martian terrain.
- Hybrid achieved 99.5% success at 10 m (vs 79.5% standalone-PPO) and 90% vs 2% at 50 m; built the A* command term, terrain importer, reward tuning, and a custom evaluation pipeline.

Tradie Text — Micro-SaaS · in production

- Built a production micro-SaaS that recovers missed calls for trades businesses using Twilio telephony and a GPT-4o-mini SMS conversation engine.
- Built multi-tenant infrastructure end-to-end: Next.js 15 + TypeScript, Supabase Postgres with row-level security, Redis-backed idempotency/rate limiting, and Stripe usage-based billing.
- Hardened the system solo with 276 CI tests, Playwright e2e coverage, a threat model, and incident runbooks.

FUN.D — Full-stack web app

- Built a full-stack crowdfunding platform: a Flask REST API (JWT auth, role-based admin) and a React 19 + Vite frontend.
- Implemented an idempotent Stripe Elements payment flow end-to-end and resolved N+1 queries, upload validation, and CI across both stacks.

Skills

Firmware & electronics: C / C++ · Arduino, FPGA · Verilog / SystemVerilog, Raspberry Pi · microcontrollers, Sensor integration & calibration, Power wiring & junction boxes, Assembly

Robotics & simulation: ROS2, Isaac Sim / Isaac Lab, Sensor fusion & path planning, HIL simulation, CAD (Fusion/SW), Mechanism design

Control & signals: Advanced control systems, Signal processing & filtering, State estimation, GNC fundamentals, MATLAB, Radar & antenna coursework

AI & computer vision: RL (PPO), skrl · PyTorch, Computer vision · MediaPipe, LLM integration (OpenAI), Evaluation pipelines, AI-native / agentic workflows

Software engineering: Python / TS, React · Next.js · Flask, Postgres · Supabase · Redis, Stripe · Twilio integration, pytest · Vitest · Playwright · CI, git · GitHub · profiling

Education

B.E. (Honours), Mechatronic Engineering

University of Sydney · 2022 — 2026

EWAM (Engineering Weighted Average Mark) 76% — Distinction